CHAPTER 6

Stickelberger's Theorem

The aim of this chapter is to give, for any abelian number field, elements of the group ring of the Galois group which annihilate the ideal class group. They will form the Stickelberger ideal. The proof involves factoring Gauss sums as products of prime ideals, and since Gauss sums generate principal ideals, we obtain relations in the ideal class group. As an application, we prove Herbrand's theorem which relates the nontriviality of certain parts of the ideal class group of $\mathbb{Q}(\zeta_p)$ to p dividing corresponding Bernoulli numbers. Then we calculate the index of the Stickelberger ideal in the group ring for $\mathbb{Q}(\zeta_{p^n})$ and find it equals the relative class number. Finally, we prove a result, essentially due to Eichler, on the first case of Fermat's Last Theorem. In the next chapter we shall use Stickelberger elements to give Iwasawa's construction of p-adic L-functions.

§6.1. Gauss Sums

In order to prove Stickelberger's theorem, we need to study Gauss sums, which are also interesting in their own right. The Gauss sums used here are not the same as those used earlier, but there are many similarities.

Let $\mathbb{F} = \mathbb{F}_q$ be the finite field with q elements, q being a power of the prime p. Let ζ_p be a fixed primitive pth root of unity and let T be the trace from $\mathbb{F}$ to $\mathbb{Z}/p\mathbb{Z}$. Define

$$\psi: \mathbb{F} \to \mathbb{C}^\times, \qquad \psi(x) = \zeta_p^{T(x)},$$

which is easily seen to be a well-defined, nontrivial (T is surjective) character of the additive group of $\mathbb{F}$. Let

$$\chi\colon \mathbb{F}^\times \to \mathbb{C}^\times$$

be a multiplicative character of $\mathbb{F}^\times$. We extend χ to all of $\mathbb{F}$ by setting $\chi(0) = 0$ (even if χ is the trivial character). Note that χ^{q-1} is the trivial character, so the order of χ is prime to p. If $q \neq p$, such characters are not the Dirichlet characters studied earlier. The concept of conductor will not enter into the present discussion.

Define the Gauss sum

$$g(\chi) = -\sum_{a \in \mathbb{F}} \chi(a)\psi(a)$$

If χ has order m, then $g(\chi) \in \mathbb{Q}(\zeta_{mp})$. A quick calculation shows that $g(1) = 1$.

Lemma 6.1. (a) $g(\bar{\chi}) = \chi(-1)\overline{g(\chi)}$;
(b) *if* $\chi \neq 1$, $g(\chi)g(\bar{\chi}) = \chi(-1)q$;
(c) *if* $\chi \neq 1$, $g(\chi)\overline{g(\chi)} = q$.

Proof. (a) is straightforward. (b) follows from (a) and (c). For (c),

$$\begin{aligned} g(\chi)\overline{g(\chi)} &= \sum_{a,b \neq 0} \chi(ab^{-1})\psi(a-b) \\ &= \sum_{b,c \neq 0} \chi(c)\psi(bc-b) \quad (\textit{let } c = ab^{-1}) \\ &= \sum_{b \neq 0} \chi(1)\psi(0) + \sum_{c \neq 0,1} \chi(c) \sum_{b \neq 0} \psi(b(c-1)) \\ &= (q-1) + \sum_{c \neq 0,1} \chi(c)(-1) = q. \end{aligned}$$

This completes the proof. □

If χ_1, χ_2 are two multiplicative characters, we define the Jacobi sum

$$J(\chi_1, \chi_2) = -\sum_{a \in \mathbb{F}} \chi_1(a)\chi_2(1-a).$$

More generally, we have

$$J(\chi_1, \ldots, \chi_n) = (-1)^{n-1} \sum_{a_1 + \cdots + a_n = 1} \chi_1(a_1) \ldots \chi_n(a_n),$$

but we do not need this for $n > 2$. Note that if χ_1 and χ_2 have orders dividing m then $J(\chi_1, \chi_2)$ is an algebraic integer in $\mathbb{Q}(\zeta_m)$.

Lemma 6.2. (a) $J(1, 1) = 2 - q$;
(b) $J(1, \chi) = J(\chi, 1) = 1$ *if* $\chi \neq 1$;
(c) $J(\chi, \bar{\chi}) = \chi(-1)$ *if* $\chi \neq 1$;
(d) $J(\chi_1, \chi_2) = g(\chi_1)g(\chi_2)/g(\chi_1\chi_2)$ *if* $\chi_1\chi_2 \neq 1$.

Proof. (a) and (b) are easy. To prove (c) and (d), we compute

$$\begin{aligned} g(\chi_1)g(\chi_2) &= \sum_{a,b} \chi_1(a)\chi_2(b)\psi(a+b) \\ &= \sum_{a,b} \chi_1(a)\chi_2(b-a)\psi(b) \\ &= \sum_{\substack{a,b \\ b \neq 0}} \chi_1(a)\chi_2(b-a)\psi(b) + \sum_a \chi_1(a)\chi_2(-a). \end{aligned}$$

If $\chi_1\chi_2 \neq 1$, then the second sum vanishes. If $\chi_1\chi_2 = 1$, then it equals $\chi_1(-1)(q-1)$. The first sum equals (let $a = bc$)

$$\sum_{\substack{b,c \\ b \neq 0}} \chi_1(b)\chi_2(b)\chi_1(c)\chi_2(1-c)\psi(b) = g(\chi_1\chi_2)J(\chi_1, \chi_2).$$

If $\chi_1\chi_2 \neq 1$, we obtain (d). If $\chi_1\chi_2 = 1$, use Lemma 6.1(b), along with $g(1) = 1$, to obtain (c). This completes the proof. □

Corollary 6.3. *If χ_1, χ_2 are characters of orders dividing m, then*

$$\frac{g(\chi_1)g(\chi_2)}{g(\chi_1\chi_2)}$$

is an algebraic integer in $\mathbb{Q}(\zeta_m)$.

Proof. If $\chi_1\chi_2$ is nontrivial, use the above result. The remaining cases are quickly checked individually. □

The significance of this result is twofold: not only is the expression integral, is also eliminates ζ_p. This will be useful later.

Let m be an integer with $(m, p) = 1$. Then the fields $\mathbb{Q}(\zeta_m)$ and $\mathbb{Q}(\zeta_p)$ are disjoint. Let $(b, m) = 1$. We may define $\sigma_b \in \mathrm{Gal}(\mathbb{Q}(\zeta_m, \zeta_p)/\mathbb{Q})$ by

$$\sigma_b\colon \zeta_p \mapsto \zeta_p, \qquad \zeta_m \mapsto \zeta_m^b.$$

(perhaps it would be better to use double indices and call this $\sigma_{1,b}$, but usually ζ_p will drop out early, leaving only ζ_m).

Lemma 6.4. *Assume χ^m is trivial. Then*

$$\frac{g(\chi)^b}{g(\chi)^{\sigma_b}} = g(\chi)^{b-\sigma_b} \in \mathbb{Q}(\zeta_m),$$

and $g(\chi)^m \in \mathbb{Q}(\zeta_m)$.

Proof. The second follows from the first if we let $b = 1 + m$. For the first, we have

$$g(\chi)^{\sigma_b} = -\sum \chi(a)^b \psi(a) = g(\chi^b).$$

Let $\tau \in \mathrm{Gal}(\mathbb{Q}(\zeta_{mp})/\mathbb{Q}(\zeta_m))$, so $\tau\colon \zeta_m \mapsto \zeta_m, \zeta_p \mapsto \zeta_p^c$ for some c, $(c, p) = 1$. Then

$$\begin{aligned} g(\chi)^\tau &= -\sum \chi(a)\psi(ca) \\ &= -\chi(c)^{-1} \sum \chi(a)\psi(a) = \chi(c)^{-1} g(\chi), \end{aligned}$$

and similarly

$$g(\chi^b)^\tau = \chi(c)^{-b} g(\chi^b).$$

Therefore τ fixes $g(\chi)^{b-\sigma_b}$. The result follows. $\square$

Lemma 6.5. $g(\chi^p) = g(\chi)$.

Proof. Since $a \mapsto a^p$ is an automorphism over $\mathbb{Z}/p\mathbb{Z}$, $T(a) = T(a^p)$, and a^p yields a permutation of $\mathbb{F}$. Therefore

$$\begin{aligned} g(\chi^p) &= -\sum \chi(a^p)\zeta_p^{T(a)} \\ &= -\sum \chi(a^p)\zeta_p^{T(a^p)} = g(\chi). \end{aligned}$$ $\square$

This completes our list of basic properties of Gauss sums. We now digress to give an application to the Fermat curve.

We wish to count the number of solutions of

$$X^d + Y^d = 1, \quad \text{with } X, Y \in \mathbb{F}_q.$$

As is usually the case, it is more natural to count points in projective space. That is, we consider solutions, except $(0, 0, 0)$, of

$$X^d + Y^d = Z^d,$$

and identify two solutions if they differ by a scalar multiple. If $Z \neq 0$, we may identify (X, Y, Z) with $(X/Z, Y/Z, 1)$ and obtain a solution of the original equation. But if $Z = 0$, we obtain the "points at infinity" $(X/0 = \infty)$, which correspond to solutions of $X^d + Y^d = 0$. Since we do not count $(0, 0, 0)$, we must have $Y \neq 0$, so any solution $(X, Y, 0)$ may be put in the form $(X/Y, 1, 0)$. The number of points at infinity is exactly the number of solutions in $\mathbb{F}_q$ of $X^d = -1$.

Despite all this, we shall start by counting the solutions of $X^d + Y^d = 1$, and make the correction later. We first assume d divides $q - 1$. Since $\mathbb{F}_q^\times$ is cyclic of order $q - 1$, there exists a character χ of $\mathbb{F}_q^\times$ of order exactly d. The cyclicity implies that $\chi(u) = 1$ if and only if $u \in \mathbb{F}_q^\times$ is a dth power. For $u \in \mathbb{F}_q^\times$, let $N_d(u)$ be the number of solutions in $\mathbb{F}_q$ of $X^d = u$, so

$$N_d(u) = \begin{cases} 1, & u = 0 \\ 0, & u \neq 0, u \neq d\text{th power} \\ d, & u \neq 0, u = d\text{th power (since } d \mid q-1). \end{cases}$$

It follows easily that

$$N_d(u) = \sum_{a=1}^{d} \chi^a(u) \quad \text{if } u \neq 0.$$

Therefore, the number of solutions of $X^d + Y^d = 1$ is

$$\sum_{\substack{u+v=1 \\ uv \neq 0}} N_d(u)N_d(v) + 2d$$

(the second term corresponds to $X = 0$ or $Y = 0$)

$$= 2d + \sum_{u \neq 0, 1} \sum_{a=1}^{d} \sum_{b=1}^{d} \chi^a(u)\chi^b(1-u)$$

$$= 2d - \sum_{a=1}^{d} \sum_{b=1}^{d} J(\chi^a, \chi^b).$$

From Lemma 6.2 we see that the term $a = b = d$ contributes $2 - q$; the terms with either $a = d$ or $b = d$, but not both, contribute a total of $2(d-1)$; those with $a + b = d$ yield $\sum_{a=1}^{d-1} \chi^a(-1) = N_d(-1) - 1$; and the remaining terms can be expressed in terms of Gauss sums via Lemma 6.2(d). Using the fact that $N_d(-1)$ is the number of points at infinity, we find that the number of solutions of $X^d + Y^d = Z^d$ in projective space is

$$q + 1 - \sum_{\substack{a,b=1 \\ a+b \neq d}}^{d-1} J(\chi^a, \chi^b).$$

Since $|g(\chi)| = \sqrt{q}$ if $\chi \neq 1$, we have, using Lemma 6.2(d), that

$$|\sum J(\chi^a, \chi^b)| \leq (d-1)(d-2)\sqrt{q}.$$

If $\bar{N}$ denotes the number of solutions,

$$|\bar{N} - (q+1)| \leq (d-1)(d-2)\sqrt{q}.$$

This is a special case of a more general result which states that for a curve of genus g we have

$$|\bar{N} - (q+1)| \leq 2g\sqrt{q}.$$

Note that $q + 1$ is the number of points on a line $aX + bY = cZ$, so the number of points on a curve is approximately the same as for a line, the possible error being bounded in terms of the genus.

Now assume d is arbitrary, so we do not necessarily have $d | q - 1$. Let $e = (d, q-1)$. Then $N_d(u) = N_e(u)$, so

$$\begin{aligned} \bar{N} &= \#\{X^d + Y^d = Z^d | (X,Y,Z) \neq (0,0,0)\}/(q-1) \\ &= \sum_{u+v=w} N_d(u)N_d(v)N_d(w)/(q-1) \\ &= \#\{X^e + Y^e = Z^e | (X,Y,Z) \neq (0,0,0)\}/(q-1). \end{aligned}$$

Since e divides $q - 1$, we have

$$|\bar{N} - (q+1)| \leq (e-1)(e-2)\sqrt{q} \leq (d-1)(d-2)\sqrt{q},$$

so we have proved the following.

Proposition 6.6. *Let $\bar{N}$ denote the number of projective space solutions of*

$$X^d + Y^d = Z^d$$

in $\mathbb{F}_q$. *Then*

$$|\bar{N} - (q + 1)| \leq (d - 1)(d - 2)\sqrt{q}. \qquad \square$$

Corollary 6.7. *For any given* d, $X^d + Y^d \equiv 1 \pmod p$ *has solutions with* $XY \not\equiv 0 \pmod p$, *for all sufficiently large* p.

Proof. From the above, the number of points at infinity is $N_d(-1)$ (or $N_e(-1)$), which is at most d. The number of solutions with $X \equiv 0$ or $Y \equiv 0$ is at most $2d$. Therefore we have a nontrivial solution as soon as $\bar{N} > 3d$. Since $\bar{N} - p = O(\sqrt{p})$, the result follows. $\square$

This corollary shows that it would be difficult to prove Fermat's Last Theorem using only congruences.

To finish this digression we show that Proposition 6.6 is essentially the Riemann hypothesis for the Fermat curve. Fix d and p and let $\bar{N}_n$ be the number of solutions of $X^d + Y^d = Z^d$ in projective space over $\mathbb{F}_{p^n}$. The zeta function $\zeta_d(s)$ of the curve may be defined as follows:

Define $Z(T)$ by

$$\frac{Z'(T)}{Z(T)} = \sum_{n=1}^{\infty} \bar{N}_n T^{n-1}, \qquad Z(0) = 1.$$

Then

$$\zeta_d(s) = Z(p^{-s}).$$

This function satisfies many properties similar to those for Dedekind zeta functions: for example, there is an Euler product, and also there is a functional equation relating the values at s and $1 - s$. It can be shown that $Z(T)$ is a rational function of the form

$$Z(T) = \frac{P(T)}{(1 - T)(1 - pT)}, \quad \text{where } P(T) \in \mathbb{Z}[T],\ P(0) = 1$$

(for further properties and proofs, see Weil [6] or Eichler [3]).

Writing $P(T) = \prod_i (1 - \alpha_i T)$, we see that

$$\frac{Z'(T)}{Z(T)} = \sum_{n=1}^{\infty} \left(1 + p^n - \sum_i \alpha_i^n\right) T^{n-1}.$$

so

$$\bar{N}_n = 1 + p^n - \sum_i \alpha_i^n.$$

To answer the question that arises when one compares this with a previous formula, yes, the α_i's are Jacobi sums, but we shall not prove this here. It follows easily from the Davenport–Hasse relations (see the Exercises).

From Proposition 6.6 we have

$$\left|\sum_i \alpha_i^n\right| \leq (d - 1)(d - 2)p^{n/2}.$$

Lemma 6.8.

$$\limsup_{n\to\infty}\left|\sum_i \alpha_i^n\right|^{1/n} = \operatorname*{Max}_i |\alpha_i|.$$

Proof. The only problem arises when two α's have the same absolute value, in which case a straightforward proof would have to show that "cancellation" does not decrease the lim sup. However, there is the following classical trick. Consider the complex function

$$f(z) = \sum_i \frac{1}{1-\alpha_i z} = \sum_{n=0}^{\infty}\left(\sum_i \alpha_i^n\right) z^n.$$

The radius of convergence of the power series is the distance to the nearest singularity, namely $1/\mathrm{Max}|\alpha_i|$. But it is also the reciprocal of $\limsup|\sum \alpha_i^n|^{1/n}$. The result follows. □

We now have $|\alpha_i| \le \sqrt{p}$ for each i. Returning to $\zeta_d(s)$, we see that $\zeta_d(s) = 0 \Leftrightarrow p^s = \alpha_i$ for some i. Therefore $\mathrm{Re}(s) \le \frac{1}{2}$. But the functional equation for $\zeta_d(s)$ implies that if $\zeta_d(s) = 0$ then $\zeta_d(1-s) = 0$. Therefore $\mathrm{Re}(s) = \frac{1}{2}$ for each zero s. This is the Riemann hypothesis for the Fermat curve.

All of the above is part of a much more general situation, which applies not only to curves but also to higher dimensional varieties (the Weil conjectures, now Deligne's theorem). The reader is strongly urged to read the classic papers of Weil ([1], [2]), where this is discussed and where additional results on Gauss and Jacobi sums are proved.

§6.2. Stickelberger's Theorem

Let $M/\mathbb{Q}$ be a finite abelian extension, so $M \subseteq \mathbb{Q}(\zeta_m)$ for some m (by the Kronecker–Weber theorem, proved in Chapter 14). We assume m is minimal. $G = \mathrm{Gal}(M/\mathbb{Q})$ may be regarded as a quotient of $(\mathbb{Z}/m\mathbb{Z})^\times$. We let σ_a, $(a,m) = 1$, denote both the element of $\mathrm{Gal}(\mathbb{Q}(\zeta_m)/\mathbb{Q})$ and its restriction to M. Let $\{x\}$ denote the fractional part of the real number x; so $x - \{x\} \in \mathbb{Z}$ and $0 \le \{x\} < 1$. Define the *Stickelberger element*

$$\theta = \theta(M) = \sum_{\substack{a \,(\mathrm{mod}\, m)\\ (a,m)=1}} \left\{\frac{a}{m}\right\}\sigma_a^{-1} \in \mathbb{Q}[G].$$

The *Stickelberger ideal* $I(M)$ is defined to be $\mathbb{Z}[G] \cap \theta\mathbb{Z}[G]$, in other words, those $\mathbb{Z}[G]$-multiples of θ which have integral coefficients.

Lemma 6.9. *Suppose $M = \mathbb{Q}(\zeta_m)$. Let I' be the ideal of $\mathbb{Z}[G]$ generated by elements of the form $c - \sigma_c$, with $(c,m) = 1$. Let $\beta \in \mathbb{Z}[G]$. Then*

$$\beta\theta \in \mathbb{Z}[G] \Leftrightarrow \beta \in I'.$$

Therefore $I = I'\theta$.

Proof. Since

$$(c-\sigma_c)\theta=\sum_a\left(c\left\{\frac{a}{m}\right\}-\left\{\frac{ac}{m}\right\}\right)\sigma_a^{-1}\in\mathbb{Z}[G],$$

we have "$\Leftarrow$". To prove the converse, first note that $m=(1+m)-\sigma_{1+m}\in I'$. Suppose $(\sum_a x_a\sigma_a)\theta\in\mathbb{Z}[G]$, with $x_a\in\mathbb{Z}$. A short calculation shows that

$$\left(\sum_a x_a\sigma_a\right)\left(\sum_c\left\{\frac{c}{m}\right\}\sigma_c^{-1}\right)=\sum_b\left(\sum_a x_a\left\{\frac{ab}{m}\right\}\right)\sigma_b^{-1}.$$

Looking at the coefficient of σ_1^{-1}, we find that m divides $\sum x_a a$. Since $m\in I'$, so is $\sum x_a a$. Therefore

$$\sum x_a\sigma_a=\sum x_a(\sigma_a-a)+\sum x_a a\in I'.$$

This completes the proof. □

This result is not true in general if M is a proper subfield of $\mathbb{Q}(\zeta_m)$. The problem is that $\sigma_b=1$ for several b, so the "coefficient of σ_1^{-1}" involves a sum over various values of b. For example, let $M=\mathbb{Q}(\sqrt{12})=\mathbb{Q}(\zeta_{12}+\zeta_{12}^{11})\subset\mathbb{Q}(\zeta_{12})$. Then $\sigma_1=\sigma_{11}=1$, while $\sigma_5=\sigma_7=\sigma$, say. We have $\theta(M)=1+\sigma\in\mathbb{Z}[G]$, so $1\cdot\theta\in\mathbb{Z}[G]$. But I' is generated by $\{5-\sigma,7-\sigma,11-1\}$, therefore by $\{2,1+\sigma\}$. In particular, $1\notin I'$.

If $x=\sum x_\sigma\sigma\in\mathbb{Z}[G]$ then x acts on ideals and ideal classes in the natural way: $A^x=\prod_\sigma(A^\sigma)^{x_\sigma}$.

Theorem 6.10 (Stickelberger's Theorem). *Let A be a fractional ideal of M, let $\beta\in\mathbb{Z}[G]$, and suppose $\beta\theta\in\mathbb{Z}[G]$. Then $A^{\beta\theta}$ is principal. Therefore, the Stickelberger ideal annihilates the ideal class group of M.*

Before starting the proof, we give two examples.

(a) Suppose M is real. Then $\sigma_a=\sigma_{-a}$ and $\{a/m\}+\{-a/m\}=1$, so

$$\theta(M)=\frac{1}{2}\sum_{a(\text{mod }m)}\sigma_a=\frac{\phi(m)}{2\deg M}\,\mathrm{Norm}_{M/\mathbb{Q}}.$$

In this case we find that the norm, or some multiple of it, annihilates the ideal class group. This of course is already obvious, since $\mathbb{Q}$ has class number one. We therefore can obtain nontrivial results only if we look at imaginary fields.

(b) Suppose $M=\mathbb{Q}(\sqrt{-m})$ is imaginary quadratic. $\mathrm{Gal}(M/\mathbb{Q})=\{1,\sigma\}$, where σ is complex conjugation. Since $A^{1+\sigma}$ is an ideal of $\mathbb{Q}$, hence principal, σ acts by inversion on the ideal class group. Let $\beta=m$. Then $\beta\theta=\sum a\sigma_a^{-1}\in\mathbb{Z}[G]$, and in the ideal class group $\beta\theta$ acts as $\sum a\chi(a)$, where χ is the quadratic character for M. We find that $\sum a\chi(a)=mB_{1,\chi}$ annihilates the ideal class group. Of course, the class number formula implies that this number is just $-mh$ (if $m>4$), so what we have is a weak form of the analytic class number formula; however, it will be proved algebraically.

More generally, let F be any totally real number field and M/F a finite abelian extension. Let $G = \mathrm{Gal}(M/F)$. Via the Artin map $A \mapsto \sigma_A \in G$, one can define partial zeta functions for $\sigma \in G$:

$$\zeta_F(\sigma, s) = \sum_{\sigma_A = \sigma} \frac{1}{NA^s} \qquad (\mathrm{Re}(s) > 1).$$

These may be meromorphically continued to the whole complex plane, and the values $\zeta_F(\sigma, -n)$ are rational numbers for $n \geq 0$. Define

$$\theta_n(M/F) = \sum_{\sigma \in G} \zeta_F(\sigma, -n)\sigma^{-1},$$

and let $I_n(M/F)$ be the ideal generated by elements of the form

$$(NA^{n+1} - \sigma_A)\theta_n(M/F),$$

where A ranges over a set of integral ideals of F not divisible by a certain finite set of prime ideals. Then $I_n(M/F)$ should annihilate some natural object, perhaps a K-group. For example, $I_1(M/\mathbb{Q})$ annihilates $K_2\mathcal{O}_M$, except possibly for the 2-part, where $\mathcal{O}_M$ is the ring of integers of M. When $M = \mathbb{Q}(\zeta_m)$ and $F = \mathbb{Q}$, we have

$$\zeta(\sigma_a, s) = \sum_{\substack{b \equiv a(m) \\ b > 0}} \frac{1}{b^s},$$

which is essentially a Hurwitz zeta function. If $n = 0$, we have

$$\theta_0(M/\mathbb{Q}) = \sum_{c (\mathrm{mod}\ m)} \left(\frac{1}{2} - \left\{\frac{c}{m}\right\}\right)\sigma_c^{-1},$$

which differs from θ by half the norm (in fact, we shall need θ_0 later). Since $K_0\mathcal{O}_M$ is essentially the ideal class group of M, the above may be regarded as an appropriate generalization of Stickelberger's Theorem. For details, see Coates [7].

We are now ready to start the proof of Stickelberger's theorem. The major step will be the factorization of certain Gauss sums. Let p be a prime and let $q = p^f$ be a power of p. Let $\mathfrak{p}$ be a prime ideal of $\mathbb{Q}(\zeta_{q-1})$ lying above p. Since $\mathbb{Z}[\zeta_{q-1}] \bmod \mathfrak{p}$ is the finite field with q elements (f = residue class degree by Theorem 2.13), and since the $(q-1)$st roots of unity are distinct mod $\mathfrak{p}$, there is an isomorphism

$$\omega = \omega_{\mathfrak{p}}\colon \mathbb{F}_q^\times \to (q-1)\text{st roots of } 1$$

satisfying

$$\omega(a) \bmod \mathfrak{p} = a \in \mathbb{F}_q^\times.$$

This ω is essentially a generalization of the ω of the previous chapter. Let $\tilde{\mathscr{P}}$ be the prime of $\mathbb{Q}(\zeta_{q-1}, \zeta_p)$ lying above $\mathfrak{p}$. For $\alpha \in \mathbb{Z}$, let $s(\alpha) = v_{\tilde{\mathscr{P}}}(g(\omega^{-\alpha}))$, where $v_{\tilde{\mathscr{P}}}$ is the $\tilde{\mathscr{P}}$-adic valuation. Clearly $s(\alpha)$ depends only on $\alpha \pmod{q-1}$.

Lemma 6.11. (a) $s(0) = 0$;
(b) $0 \le s(\alpha + \beta) \le s(\alpha) + s(\beta)$;
(c) $s(\alpha + \beta) \equiv s(\alpha) + s(\beta) \pmod{p-1}$;
(d) $s(p\alpha) = s(\alpha)$;
(e) $\sum_{\alpha=1}^{q-2} s(\alpha) = (q-2)(f)(p-1)/2$.

Proof. (a) is obvious; (b) and (d) follow from Corollary 6.3 and Lemma 6.5, respectively. Since $\tilde{\mathscr{P}}^{p-1} = \mathfrak{p}$, the values of $v_{\tilde{\mathscr{P}}}$ on $\mathbb{Q}(\zeta_{q-1})$ are divisible by $p-1$. Therefore (c) also follows from Corollary 6.3. Since $g(\omega^{-\alpha})g(\omega^{\alpha}) = \pm q = \pm p^f$, we have $s(\alpha) + s(q-1-\alpha) = v_{\tilde{\mathscr{P}}}(p^f) = (p-1)f$. Pairing up the terms in the sum (the term for $\alpha = (q-1)/2$ pairs with itself), we obtain (e). This completes the proof. $\square$

Lemma 6.12. *$s(\alpha) > 0$ if $\alpha \not\equiv 0 \pmod{q-1}$, and $s(1) = 1$.*

Proof. Since $\pi = \zeta_p - 1 \in \tilde{\mathscr{P}}$,

$$g(\omega^{-\alpha}) = -\sum \omega^{-\alpha}(a)\zeta_p^{T(a)} \equiv -\sum \omega^{-\alpha}(a) \equiv 0 \pmod{\tilde{\mathscr{P}}}.$$

Therefore $s(\alpha) > 0$. Also,

$$\begin{aligned} g(\omega^{-1}) &= -\sum \omega^{-1}(a)\zeta_p^{T(a)} \\ &= -\sum \omega^{-1}(a)(1+\pi)^{T(a)} \equiv -\sum \omega^{-1}(a)(1+\pi T(a)) \pmod{\tilde{\mathscr{P}}^2} \\ &\equiv -\pi \sum \omega^{-1}(a)T(a). \end{aligned}$$

Regarding $\mathbb{F}_q$ as $\mathbb{Z}[\zeta_{q-1}] \bmod \mathfrak{p}$, we have

$$T(a) = a + a^p + \cdots + a^{p^{f-1}} \pmod{\mathfrak{p}}$$

(since $a \mapsto a^p$ generates the Galois group mod $\mathfrak{p}$), and

$$\sum \omega^{-1}(a)T(a) \equiv \sum_{\substack{a \not\equiv 0 \\ a \bmod \mathfrak{p}}} a^{-1}(a + a^p + \cdots + a^{p^{f-1}}) \pmod{\mathfrak{p}}.$$

If $0 < b < f$, then $\sum_{a \not\equiv 0} a^{p^b-1} \equiv 0 \pmod{\mathfrak{p}}$, so the sum reduces to $\sum_{a \not\equiv 0} 1 = q - 1 \equiv -1$. Therefore

$$g(\omega^{-1}) \equiv \pi \pmod{\tilde{\mathscr{P}}^2},$$

hence $s(1) = v_{\tilde{\mathscr{P}}}(\pi) = 1$ (since $\mathbb{Q}(\zeta_{q-1}, \zeta_p)/\mathbb{Q}(\zeta_p)$ is unramified at p). This completes the proof. $\square$

Proposition 6.13. *Let $0 \le \alpha < q - 1$ and let $\alpha = a_0 + a_1 p + \cdots + a_{f-1}p^{f-1}$, $0 \le a_i \le p-1$, be the standard p-adic expansion of α. Then*

$$s(\alpha) = a_0 + a_1 + \cdots + a_{f-1}.$$

Proof. From Lemma 6.11(a), (b), (c) and Lemma 6.12 we immediately have $s(\alpha) = \alpha$ for $0 \le \alpha \le p - 2$. If $q = p$, we are done. Otherwise, $s(p-1) > 0$

and we similarly obtain $s(p-1) = p-1$. Lemma 6.11(b) and (d) imply that $s(\alpha) \le a_0 + \cdots + a_{f-1}$. When α runs through the integers from 0 to $q-1$, inclusive, each coefficient of the p-adic expansion takes on each of the values from 0 to $p-1$ exactly p^{f-1} times, so

$$\sum_{\alpha=0}^{q-1} (a_0 + \cdots + a_{f-1}) = \frac{p(p-1)}{2}(f)p^{f-1} = \frac{p-1}{2}fq.$$

If we omit $\alpha = q - 1 = (p-1) + \cdots + (p-1)p^{f-1}$, we obtain

$$\sum_{\alpha=0}^{q-2} (a_0 + \cdots + a_{f-1}) = \frac{p-1}{2}fq - (p-1)f = \sum_{\alpha=0}^{q-2} s(\alpha),$$

by Lemma 6.11(e). The result follows. $\square$

Remark. Let $\pi = \zeta_p - 1$ and $0 \le \alpha < q-1$, as above. Then

$$g(\omega^{-\alpha}) \equiv \frac{\pi^{a_0+\cdots+a_{f-1}}}{(a_0!)\dots(a_{f-1}!)} \pmod{\tilde{\mathcal{P}}^{a_0+\cdots+a_{f-1}+1}}.$$

(see Lang [4], [5]). In Lemma 6.12 we verified a special case of this formula. The general argument follows a similar line, but involves a rather delicate analysis of binomial coefficients.

Now fix a positive integer m. Let p be a prime, $(p,m) = 1$, and let f be the order of $p \pmod{m}$, so m divides $p^f - 1 = q - 1$. Fix a prime $\wp_0$ of $\mathbb{Q}(\zeta_m)$ lying above p; let $\tilde{\wp}_0$ be the prime of $\mathbb{Q}(\zeta_m, \zeta_p)$ above $\wp_0$, so $\tilde{\wp}_0^{p-1} = \wp_0$; let $\mathcal{P}_0$ be a prime of $\mathbb{Q}(\zeta_{q-1})$ lying above $\wp_0$; and let $\tilde{\mathcal{P}}_0$ be the prime of $\mathbb{Q}(\zeta_{q-1}, \zeta_p)$ lying above $\mathcal{P}_0$ (and $\tilde{\wp}_0$). Let $\omega = \omega_{\mathcal{P}_0}$ be as above and let $\chi = \omega^{-d}$, where $d = (q-1)/m$. Then $\chi^m = 1$, so $g(\chi) \in \mathbb{Q}(\zeta_m, \zeta_p)$. Since $g(\chi)\overline{g(\chi)} = q = p^f$, the factorization of $g(\chi)$ involves only primes of $\mathbb{Q}(\zeta_m, \zeta_p)$ above p, that is, the conjugates over $\mathbb{Q}$ of $\tilde{\wp}_0$. Let $(a,m) = 1$ and let $\sigma_a \in \mathrm{Gal}(\mathbb{Q}(\zeta_m)/\mathbb{Q})$ be the corresponding element of the Galois group. For each such a, fix an extension of σ_a to $\mathbb{Q}(\zeta_{q-1}, \zeta_p)$ such that $\zeta_p^{\sigma_a} = \zeta_p$.

$$\begin{tikzcd}
 & \mathbb{Q}(\zeta_{q-1},\zeta_p)\ \tilde{\mathcal{P}}_0 \arrow[dl, dash] \arrow[dr, dash] & \\
\mathbb{Q}(\zeta_{q-1})\ \wp_0 \arrow[dr, dash] & & \mathbb{Q}(\zeta_m,\zeta_p)\ \tilde{\wp}_0 \arrow[dl, dash] \\
 & \mathbb{Q}(\zeta_m)\ \wp_0 \arrow[d, dash] & \\
 & \mathbb{Q}\ p &
\end{tikzcd}$$

The decomposition group for p in $(\mathbb{Z}/m\mathbb{Z})^\times$ is generated by $p \pmod m$ (see the discussion after Theorem 2.13). Let R denote a set of representatives for $(\mathbb{Z}/m\mathbb{Z})^\times$ modulo this decomposition group. Then $\{\wp_0^{\sigma_a^{-1}} | a \in R\}$ is the set of conjugates of $\wp_0$. Since $\tilde{\wp}_0$ is the unique prime above $\wp_0$, all conjugates of $\tilde{\wp}_0$ have the form $\tilde{\wp}_0^{\sigma_a^{-1}}$. Let $\tilde{\wp} = \tilde{\wp}_0^{\sigma_a^{-1}}$ be one of them. Then

$$v_{\tilde{\wp}}(g(\chi)) = v_{\tilde{\wp}_0}(g(\chi)^{\sigma_a}) = v_{\tilde{\wp}_0}(g(\chi^a)) = v_{\tilde{\mathscr{P}}_0}(g(\chi^a)) = s(ad)$$

($v_{\tilde{\wp}_0} = v_{\tilde{\mathscr{P}}_0}$ since $\tilde{\mathscr{P}}_0/\tilde{\wp}_0$ is unramified). Therefore

$$(g(\chi)) = \tilde{\wp}_0^{\sum_R s(ad)\sigma_a^{-1}}.$$

Lemma 6.14. *Let* $0 \le h < q - 1$. *Then*

$$s(h) = (p-1)\sum_{i=0}^{f-1}\left\{\frac{p^i h}{q-1}\right\}.$$

Proof. Let $h = a_0 + a_1 p + \cdots + a_{f-1}p^{f-1}$. Then

$$p^i h \equiv a_0 p^i + a_1 p^{i+1} + \cdots + a_{f-1}p^{i-1} \pmod{q-1}.$$

It follows that

$$\left\{\frac{p^i h}{q-1}\right\} = \frac{1}{q-1}(a_0 p^i + \cdots + a_{f-1}p^{i-1}).$$

Summing over i, we obtain the result. □

We now have $s(ad) = (p-1)\sum_{i=0}^{f-1}\{p^i a/m\}$, so

$$\sum_R s(ad)\sigma_a^{-1} = (p-1)\sum_{i=0}^{f-1}\sum_R\left\{\frac{p^i a}{m}\right\}\sigma_a^{-1}.$$

Since $\tilde{\wp}_0^{p-1} = \wp_0$ and since $\sigma_{p^i}(\wp_0) = \wp_0$ (definition of decomposition group),

$$(g(\chi)^m) = \wp_0^{m\sum\sum\{p^i a/m\}\sigma_{ap^i}^{-1}} = \wp_0^{m\theta},$$

where $\theta = \sum_{b=1,(b,m)=1}^{m}\{b/m\}\sigma_b^{-1}$ is the Stickelberger element (we raise to the mth power to avoid denominators).

We now have a partial result: If $\wp_0$ is a prime of $\mathbb{Q}(\zeta_m)$ with $\wp_0 \nmid m$, then $\wp_0^{m\theta}$ is principal in $\mathbb{Q}(\zeta_m, \zeta_p)$. The main problem is now to get down to $\mathbb{Q}(\zeta_m)$, then to M.

Suppose that A is an ideal of $M \subseteq \mathbb{Q}(\zeta_m)$ with $(A, m) = 1$. Let $A = \prod \wp_i$ be its factorization into (not necessarily distinct) prime ideals in $\mathbb{Q}(\zeta_m)$. Then

$$A^{m\theta} = \left(\prod g(\chi_{\wp_i})^m\right),$$

where we write $\chi_{\wp_i}$ to indicate that χ depends on $\wp_i$. Suppose $\beta \in \mathbb{Z}[G]$ ($G = \mathrm{Gal}(M/\mathbb{Q})$) and $\beta\theta \in \mathbb{Z}[G]$. Extending the elements of G, we may regard $\beta\theta$ as an element of $\mathbb{Z}[\mathrm{Gal}(\mathbb{Q}(\zeta_{mp})/\mathbb{Q})]$. Then

$$A^{m\beta\theta} = (\gamma^{\beta m}), \quad \text{where} \quad \gamma = \prod g(\chi_{\wp_i}) \in \mathbb{Q}(\zeta_{pm}),$$

where $P = \prod p_i$ is the product of the rational primes divisible by the $\not{p}_i$'s. Since $\gamma^{m\beta} \in \mathbb{Q}(\zeta_m)$ by Lemma 6.4, and it is the mth power of an ideal of $\mathbb{Q}(\zeta_m)$, namely $A^{\beta\theta}$, it follows that the extension $\mathbb{Q}(\zeta_m, \gamma^\beta)/\mathbb{Q}(\zeta_m)$ can be ramified only at primes dividing m (*proof*: locally, $A^{\beta\theta}$ is principal, so we are adjoining the mth root of a local unit). But

$$\mathbb{Q}(\zeta_m) \subseteq \mathbb{Q}(\zeta_m, \gamma^\beta) \subseteq \mathbb{Q}(\zeta_m, \zeta_P).$$

Therefore, ramification can occur only at p_i's. Since $(P, m) = 1$, the extension must be unramified.

Lemma 6.15. *If* $\mathbb{Q}(\zeta_m) \subseteq K \subseteq \mathbb{Q}(\zeta_n)$ *and* $K/\mathbb{Q}(\zeta_m)$ *is unramified at all primes, then* $K = \mathbb{Q}(\zeta_m)$.

Proof. Suppose $K \neq \mathbb{Q}(\zeta_m)$. Then there is a character χ for K of conductor not dividing m. By Theorem 3.5, $K/\mathbb{Q}(\zeta_m)$ must be ramified at some prime. Contradiction. For another proof, see Lemma 15.48. □

We find that $\gamma^\beta \in \mathbb{Q}(\zeta_m)$. Therefore $A^{\beta\theta} = (\gamma^\beta)$ is principal as an ideal of $\mathbb{Q}(\zeta_m)$. But this does not necessarily mean that it is principal as an ideal of M. So we show that $\gamma^\beta \in M$, which suffices, since if two ideals of M are equal in $\mathbb{Q}(\zeta_m)$ they must have been equal originally because of unique factorization.

Let $\mathscr{P}$ be a prime of $\mathbb{Q}(\zeta_{q-1})$ lying over one of the prime factors $\not{p}_i$ of A. Then $\chi_{\not{p}_i}$ *a priori* depends on the choice of $\mathscr{P}$, so we temporarily let $\chi_{\not{p}_i} = \chi_{\mathscr{P}}$. Let $\sigma \in \mathrm{Gal}(\mathbb{Q}(\zeta_{q-1})/M)$. Then

$$\sigma\colon \mathbb{Z}[\zeta_{q-1}] \bmod \mathscr{P} \xrightarrow{\sim} \mathbb{Z}[\zeta_{q-1}] \bmod \mathscr{P}^\sigma$$

and correspondingly if $\chi_{\mathscr{P}}(a) = \zeta$ then $\chi_{\mathscr{P}^\sigma}(a) = \zeta^\sigma$. Therefore $\chi_{\mathscr{P}}^\sigma = \chi_{\mathscr{P}^\sigma}$. But $\chi_{\mathscr{P}}^m = 1$, so $\chi_{\mathscr{P}^\sigma} = \chi_{\mathscr{P}}$ for $\sigma \in \mathrm{Gal}(\mathbb{Q}(\zeta_{q-1})/\mathbb{Q}(\zeta_m))$. Therefore $\chi_{\mathscr{P}}$ depends only on $\not{p}_i$, so we may return to the notation $\chi_{\not{p}_i}$. The above reasoning shows that $\chi_{\not{p}_i}^\sigma = \chi_{\not{p}_i^\sigma}$ for $\sigma \in \mathrm{Gal}(\mathbb{Q}(\zeta_m)/M)$. If we extend σ by letting $\sigma(\zeta_p) = \zeta_p$, then $g(\chi_{\not{p}_i})^\sigma = g(\chi_{\not{p}_i}^\sigma) = g(\chi_{\not{p}_i^\sigma})$.

Since $A^\sigma = A$ for $\sigma \in \mathrm{Gal}(\mathbb{Q}(\zeta_m)/M)$, σ permutes the $\not{p}_i$'s. Therefore

$$\gamma^{\beta\sigma} = \prod g(\chi_{\not{p}_i})^{\beta\sigma} = \prod g(\chi_{\not{p}_i^\sigma})^\beta = \gamma^\beta.$$

But we already have $\gamma^\beta \in \mathbb{Q}(\zeta_m)$; hence $\gamma^\beta \in M$. So $A^{\beta\theta}$ is principal in M.

Finally, if A is an arbitrary ideal of M, we may write $A = (a)A_1$, with $a \in M$ and $(A_1, m) = 1$. Then

$$A^{\beta\theta} = (a^{\beta\theta})A_1^{\beta\theta},$$

which is principal. This completes the proof of Stickelberger's theorem. □

For an easier proof of Stickelberger's theorem in the case of a full cyclotomic field $\mathbb{Q}(\zeta_m)$, see Section 15.1.

§6.3. Herbrand's Theorem

Let G be a finite abelian group and $\hat{G}$ its character group. Let $\chi \in \hat{G}$ and define

$$\varepsilon_\chi = \frac{1}{|G|} \sum_{\sigma \in G} \chi(\sigma)\sigma^{-1} \in \bar{\mathbb{Q}}[G],$$

where $\bar{\mathbb{Q}}$ is the algebraic closure of $\mathbb{Q}$. One may easily verify the following relations:

(a) $\varepsilon_\chi^2 = \varepsilon_\chi$;
(b) $\varepsilon_\chi \varepsilon_\psi = 0$ if $\chi \neq \psi$;
(c) $1 = \sum_{\chi \in \hat{G}} \varepsilon_\chi$;
(d) $\varepsilon_\chi \sigma = \chi(\sigma)\varepsilon_\chi$.

The ε_χ's are called the orthogonal idempotents of the group ring $\bar{\mathbb{Q}}[G]$. If M is a module over $\bar{\mathbb{Q}}[G]$ then we may write

$$M = \bigoplus_\chi M_\chi, \quad \text{where } M_\chi = \varepsilon_\chi M$$

(use (c) to get the sum; if $0 = \sum \varepsilon_\chi a_\chi$, then use (b) and (a) to show $\varepsilon_\chi a_\chi = 0$ for all χ). Each $\sigma \in G$ acts on M, and M_χ is the eigenspace with eigenvalue $\chi(\sigma)$, by (*d*).

Of course, all the above works if $\bar{\mathbb{Q}}$ is replaced by any (commutative) ring which contains the values of all $\chi \in \hat{G}$ and in which $|G|$ is invertible.

In particular, let p be an odd prime and let $G = \mathrm{Gal}(\mathbb{Q}(\zeta_p)/\mathbb{Q}) \simeq (\mathbb{Z}/p\mathbb{Z})^\times$. Then $\hat{G} = \{\omega^i | 0 \leq i \leq p-2\}$. We shall work in the group ring $\mathbb{Z}_p[G]$. The idempotents are

$$\varepsilon_i = \frac{1}{p-1} \sum_{a=1}^{p-1} \omega^i(a)\sigma_a^{-1}, \qquad 0 \leq i \leq p-2.$$

Later, we shall also need

$$\varepsilon_- = \frac{1 - \sigma_{-1}}{2} = \sum_{i \text{ odd}} \varepsilon_i \quad \text{and} \quad \varepsilon_+ = \frac{1 + \sigma_{-1}}{2} = \sum_{i \text{ even}} \varepsilon_i.$$

There is a decomposition $A = A^- \oplus A^+$ for any $\mathbb{Z}_p[G]$-module, for example the p-Sylow subgroup of the ideal class group.

Let $\theta = (1/p)\sum_{a=1}^{p-1} a\sigma_a^{-1}$ be the Stickelberger element. Using (d), we find that

$$\varepsilon_i \theta = \frac{1}{p} \sum_{a=1}^{p-1} a\omega^{-i}(a)\varepsilon_i = B_{1,\omega^{-i}}\varepsilon_i$$

and

$$\varepsilon_i(c - \sigma_c)\theta = (c - \omega^i(c))B_{1,\omega^{-i}}\varepsilon_i.$$

Let A be the p-Sylow subgroup of the ideal class group of $\mathbb{Q}(\zeta_p)$. Since $p^n A = 0$ for sufficiently large n, we may make A into a $\mathbb{Z}_p$-module by defining

$$\left(\sum_{j=0}^{\infty} b_j p^j\right) a = \sum_{j=0}^{\infty} (b_j p^j a),$$

since the latter sum is finite. G also acts on A, so A is a $\mathbb{Z}_p[G]$-module. Let

$$A = \bigoplus_{i=0}^{p-2} A_i$$

be the decomposition as above. Stickelberger's theorem implies that $(c - \sigma_c)\theta$ annihilates A, hence each A_i. Therefore we have proved the following: Let $c \in \mathbb{Z}$, $(c, p) = 1$. Then $(c - \omega^i(c))B_{1,\omega^{-i}}$ annihilates A_i.

Remark. Since $p\theta \equiv (p-1)\varepsilon_1 (\operatorname{mod} p)$, it is not very surprising that $p\theta$ annihilates A_i for $i \neq 1$. The fact that it annihilates A_1, however, requires Stickelberger's theorem.

Now, suppose $i \neq 0$ is even. Then $B_{1,\omega^{-i}} = 0$ so the above says nothing. If $i = 0$ then $(c-1)/2$ annihilates A_0, so $A_0 = 0$. But this is already obvious since $\varepsilon_0 = (\text{Norm})/(p-1)$.

Let i be odd. Consider first the case $i = 1$. Let $c = 1 + p$, so we have

$$\begin{aligned}(c - \omega(c))B_{1,\omega^{-1}} = pB_{1,\omega^{-1}} &= \sum_{a=1}^{p-1} a\omega^{-1}(a) \\ &\equiv p - 1 \not\equiv 0 \pmod p.\end{aligned}$$

Since A_1 is a p-group, we must have $A_1 = 0$. (It is easily seen that $A_1 = 0$ is related to von Staudt–Clausen, which is related to the fact that the p-adic zeta function has a pole. Perhaps this explains why A_1 is a special case). If $i \neq 1$, we may choose an integer c, for example a primitive root $(\operatorname{mod} p)$, such that $c \not\equiv c^i \equiv \omega^i(c) \pmod p$. We may consequently ignore the factor $c - \omega^i(c)$, so we obtain the following.

Proposition 6.16. $A_0 = A_1 = 0$. *For* $i = 3, 5, \ldots, p-2$, $B_{1,\omega^{-i}}$ *annihilates* A_i. □

Suppose $A_i \neq 0$. Then we must have $B_{1,\omega^{-i}} \equiv 0 \pmod p$. But $B_{1,\omega^{-i}} \equiv B_{p-i}/(p-i) \pmod p$ by Corollary 5.15. We have proved the following.

Theorem 6.17 (Herbrand). *Let i be odd*, $3 \leq i \leq p-2$. *If* $A_i \neq 0$ *then* $p \mid B_{p-i}$. □

This theorem is much stronger than the theorem "$p \mid h \Rightarrow p$ divides some Bernoulli number" since it gives a "piece-by-piece" description of the criterion. Even better, the following is true.

Theorem 6.18 (Ribet). *Let i be odd, $3 \le i \le p-2$. If $p | B_{p-i}$ then $A_i \ne 0$.*

This will be proved in Chapter 15 by elementary means. Ribet's original proof used delicate techniques from algebraic geometry to construct an abelian unramified extension of degree p which corresponds by class field theory to A_i.

One corollary of Ribet's theorem is that the p-rank of the ideal class group of $\mathbb{Q}(\zeta_p)$ is at least the index of irregularity (p-rank = number of summands when A is decomposed as a direct sum of cyclic groups of p-power order). However, it is not known whether or not there is equality, since the rank of some A_i could possibly be two or larger. If $p \nmid h(\mathbb{Q}(\zeta_p)^+)$ then we do have equality, as we shall prove in Chapter 10.

§6.4. The Index of the Stickelberger Ideal

Let p be an odd prime, $n \ge 1$, $G = \mathrm{Gal}(\mathbb{Q}(\zeta_{p^n})/\mathbb{Q})$, and $R = \mathbb{Z}[G]$. As before,

$$\theta = \frac{1}{p^n} \sum_{\substack{a=1 \\ (a,p)=1}}^{p^n} a\sigma_a^{-1}$$

is the Stickelberger element and $I = R\theta \cap R$ is the Stickelberger ideal. Let $J = \sigma_{-1}$ denote complex conjugation. Then

$$R^- = \{x \in R | Jx = -x\} = (1-J)R,$$

the first equality being the definition, the second following from a short calculation. We define

$$I^- = I \cap R^- = R\theta \cap R^-.$$

Note that we have to be careful about using the idempotent $(1-J)/2$ since it has a denominator. However, observe that $x \in R^- \Leftrightarrow [(1-J)/2]x = x$.

Theorem 6.19 (Iwasawa). $[R^- : I^-] = h^-(\mathbb{Q}(\zeta_{p^n}))$.

Remark. The above definitions hold for arbitrary $\mathbb{Q}(\zeta_m)$. Kučera [1] has calculated this index when it is finite. Sinnott [1] defined a larger ideal S^- (equal to I^- when m is a prime power) and showed that $[R^- : S^-]$ is a power of 2 times $h^-(\mathbb{Q}(\zeta_m))$.

Proof of Theorem 6.19. The proof will proceed by considering completions, since we can then work with one prime at a time, which is slightly easier. Let q be a prime, $R_q = \mathbb{Z}_q[G]$, $I_q = R_q I$. Clearly I is dense in I_q in the natural q-adic topology. Also, $R_q^- = (1-J)R_q$ and $I_q^- = I_q \cap R_q^-$.

Lemma 6.20. (a) $I_q = R_q\theta \cap R_q$, (b) $I_q^- = R_q\theta \cap R_q^-$, (c) $I_q^- = I^- \cdot \mathbb{Z}_q$, (d) *If $p \ne q$ then $I_q = R_q\theta$.*

Proof. The proof of Lemma 6.9 works for both R and R_q, with $I'_q = I'\mathbb{Z}_q$, so we obtain

$$R_q\theta \cap R_q = I'_q\theta = I'\mathbb{Z}_q\theta = I'\theta\mathbb{Z}_q = I\mathbb{Z}_q = I_q.$$

This proves (a). Part (b) follows easily from (a). If $p \neq q$ then $\theta \in R_q$, so (d) also follows from (a).

We now prove (c). Since $\{z\} + \{-z\} = 1$ for $z \notin \mathbb{Z}$, we have

$$(1 + J)\theta = N, \quad \text{where } N = \sum_{\sigma \in G} \sigma.$$

Let $x \in I'$. Then $x\theta \in I$, and we have

$$x\theta \in I^- \Leftrightarrow (1 + J)x\theta = 0 \Leftrightarrow xN = 0.$$

Similarly, suppose $y \in I'_q$. Then $y\theta \in I'_q\theta = I_q$, from the above, and

$$y\theta \in I_q^- \Leftrightarrow yN = 0.$$

Clearly $I^-\mathbb{Z}_q \subseteq I_q^-$. Suppose now that $y\theta \in I_q^-$, with $y \in I'_q$. We may write

$$y = \sum_c \sum_\sigma a_\sigma^c \sigma(c - \sigma_c), \qquad a_\sigma^c \in \mathbb{Z}_q.$$

The condition $yN = 0$ becomes $\sum_c \sum_\sigma a_\sigma^c(c - 1) = 0$. We want to approximate y by an element $x \in I'$ such that $xN = 0$. This then will give us an element $x\theta$ of I^- near $y\theta$, which will show that $I_q^- \subseteq$ closure of $I^- = I^-\mathbb{Z}_q$, as desired. The approximation will reduce to the following.

Fact. *Suppose $b_i \in \mathbb{Z}$, $s_i \in \mathbb{Z}_q$, and suppose $\sum_{i=1}^m b_i s_i = 0$. Then there is a sequence $(t_1^{(n)}, \ldots, t_m^{(n)}) \in \mathbb{Z}^m$ whose limit is $(s_1, \ldots, s_m)$ and such that $\sum b_i t_i^{(n)} = 0$.*

Proof. We may assume $(q, b_1) = 1$. For $2 \leq i \leq m$, choose $t_i^{(n)} \equiv 0 \bmod b_1$ with $t_i^{(n)}$ near s_i (this is where $(q, b_1) = 1$ is needed). Then $\sum b_i t_i^{(n)} \equiv 0 \bmod b_1$, so we can choose $t_1^{(n)} \in \mathbb{Z}$. Since $t_i^{(n)}$ is near s_i for $i \neq 1$, we must also have $t_1^{(n)}$ near s_1. This completes the proof of the fact.

If we let $a_\sigma^c = s_{\sigma,c}(= s_i)$, $c - 1 = b_{\sigma,c}$, and $x = \sum t_{\sigma,c}^{(n)}\sigma(c - \sigma_c) \in I'$, then x is near y and $xN = 0$, as desired. This completes the proof of Lemma 6.20. □

We have $R_q \simeq R \otimes \mathbb{Z}_q$, and under this isomorphism $R_q^- \simeq R^- \otimes \mathbb{Z}_q$ and $I_q^- \simeq I^- \otimes \mathbb{Z}_q$, by (c). It follows easily that $R_q^-/I_q^- \simeq (R^-/I^-) \otimes \mathbb{Z}_q$, which is isomorphic to the q-part of R^-/I^-. Therefore it suffices to prove the following.

Theorem 6.21. $[R_q^- : I_q^-] =$ *q-part of $h^-(\mathbb{Q}(\zeta_{p^n}))$.*

Proof. We first consider $q \neq 2$, p. Then $(1 \pm J)/2 \in R_q$; and we get $R_q = R_q^+ \oplus R_q^-$ and $I_q = I_q^+ \oplus I_q^-$ from the relation $1 = (1 + J)/2 + (1 - J)/2$. Therefore

$$I_q^- = \frac{1 - J}{2} I_q = \frac{1 - J}{2} R_q\theta = R_q^-\theta.$$

Consider the linear map

$$A\colon R_q^- \to R_q^-, \qquad x \mapsto x\theta.$$

By the theory of elementary divisors, as in the proof of Lemma 4.15, we have $[R_q^- : R_q^-\theta] = q$-part of $\det(A)$. But $\det(A)$ may be computed by working in $\bar{\mathbb{Q}}_q[G]^-$, which has the advantage of being a vector space over an algebraically closed field. We have

$$\bar{\mathbb{Q}}_q[G]^- = \bigoplus_{\chi \text{ odd}} \varepsilon_\chi \bar{\mathbb{Q}}_q[G]$$

where $\varepsilon_\chi = (1/p^n)\sum_{a=1,(a,p)=1}^{p^n} \chi(a)\sigma_a^{-1}$ and each direct summand is one-dimensional. As in the previous section,

$$\varepsilon_\chi \theta = B_{1,\bar{\chi}}\varepsilon_\chi,$$

so A becomes a diagonal matrix. Therefore $\det(A) = \prod_{\chi \text{ odd}} B_{1,\bar{\chi}}$, hence

$$\begin{aligned}[R_q^- : I_q^-] &= q\text{-part of} \prod_\chi B_{1,\bar{\chi}} \\ &= q\text{-part of } 2p^n \prod (-\tfrac{1}{2}B_{1,\bar{\chi}}) \\ &= q\text{-part of } h^-(\mathbb{Q}(\zeta_{p^n})).\end{aligned}$$

For $q = 2$, the argument must change slightly since $(1 + J)/2 \notin R_2$, so $R_2 \neq R_2^+ \oplus R_2^-$. Also, there is a power of 2 in the class number formula which must be accounted for.

Since we are restricted in our use of $(1 - J)/2$, we modify θ to obtain an element already in $\mathbb{Q}_2[G]^-$. Let

$$\tilde{\theta} = \sum_{\substack{a=1\\(a,p)=1}}^{p^n} \left(\frac{a}{p^n} - \frac{1}{2}\right)\sigma_a^{-1} = \theta - \tfrac{1}{2}N,$$

where N is the norm (one could also call it the trace). Clearly $\tilde{\theta} \notin R_2$, but a short calculation shows that

$$\frac{1 - J}{2}\tilde{\theta} = \tilde{\theta}$$

so $\tilde{\theta}$ is in the "$-$" component. We recall that $\tilde{\theta}$ is perhaps a better Stickelberger element than θ, since it is the one that generalizes most readily (see the discussion after the statement of Theorem 6.10).

Lemma 6.22. (a) $I_2^- \subseteq R_2\tilde{\theta}$;
(b) $[R_2\tilde{\theta} : I_2^-] = 2$.

Proof. For the first statement, suppose $x \in R_2$ and $x\theta \in I_2^- = R_2\theta \cap R_2^-$. Then $x\theta = [(1 - J)/2]x\theta = x[(1 - J)/2](\tilde{\theta} + \frac{1}{2}N) = x\tilde{\theta} \in R_2\tilde{\theta}$.

For (b), we claim that if $x \in R_2$ then either $x\tilde{\theta} \in R_2$ or $x\tilde{\theta} - \tilde{\theta} \in R_2$. To prove this, we note that $x\tilde{\theta} = x\theta - \frac{1}{2}xN \in R_2 \Leftrightarrow \frac{1}{2}xN \in R_2$ and similarly for

$(x-1)$. Let $x = \sum x_\sigma \sigma$. Then $xN = (\sum x_\sigma)N$ and $(x-1)N = (-1 + \sum x_\sigma)N$. Since either $(\sum x_\sigma)$ or $(-1 + \sum x_\sigma)$ is even, the claim is established.

The claim implies that $[R_2\tilde{\theta} : R_2\tilde{\theta} \cap R_2] = 2$ (the index is not 1 since $\tilde{\theta} \notin R_2$). The proof of the lemma will be complete if we can show that $R_2\tilde{\theta} \cap R_2 = R_2\theta \cap R_2^- = I_2^-$. We have already shown that $I_2^- \subseteq R_2\tilde{\theta} \cap R_2$. Now, let $x\tilde{\theta} \in R_2\tilde{\theta} \cap R_2$, where $x = \sum x_\sigma \sigma \in R_2$. Then, as above, $x\tilde{\theta} \in R_2 \Rightarrow \frac{1}{2}xN \in R_2 \Rightarrow \sum x_\sigma \equiv 0 \pmod 2$. Let $y_\sigma = x_\sigma$ for $\sigma \neq 1$, J, and let $y_1 = x_1 - \frac{1}{2}\sum x_\sigma$ and $y_J = x_J - \frac{1}{2}\sum x_\sigma$. Then $\sum y_\sigma = 0$, so $y = \sum y_\sigma \sigma \in R_2$ satisfies $yN = 0$. Also, $x - y = (\frac{1}{2}\sum x_\sigma)(1 + J)$, so $(x-y)\tilde{\theta} = 0$. Hence

$$x\tilde{\theta} = y\tilde{\theta} = y\theta - \tfrac{1}{2}yN = y\theta \in R_2\theta.$$

Since $x\tilde{\theta} \in R_2$ and satisfies $[(1-J)/2]x\tilde{\theta} = x\tilde{\theta}$, we have

$$x\tilde{\theta} \in R_2\theta \cap R_2^- = I_2^-, \quad \text{so} \quad R_2\tilde{\theta} \cap R_2 = I_2^-.$$

This completes the proof of Lemma 6.22. □

Just as for the other primes, we have a linear map

$$A : R_2^- \to R_2^-, \qquad x \mapsto \tilde{\theta}x$$

(since $x \in R_2^-$, $\frac{1}{2}xN = 0$, so there is no 2 in the denominator), and

$$\begin{aligned}[R_2^- : R_2^-\tilde{\theta}] &= \text{2-part of } \det(A) \\ &= \text{2-part of } \prod_{\chi \text{ odd}} B_{1,\bar{\chi}} \qquad (\chi \text{ odd} \Rightarrow \varepsilon_\chi\tilde{\theta} = \varepsilon_\chi\theta) \\ &= 2^{(1/2)|G|} \cdot \tfrac{1}{2} \cdot (\text{2-part of } h^-).\end{aligned}$$

Since this index is finite we must have

$$\tfrac{1}{2}|G| = \mathbb{Z}_2\text{-rank of } R_2^- = \mathbb{Z}_2\text{-rank of } R_2^-\tilde{\theta}.$$

Observe that $R_2^-\tilde{\theta} = (1-J)R_2\tilde{\theta} = R_2(2\tilde{\theta}) = 2R_2\tilde{\theta}$. Therefore

$$[R_2\tilde{\theta} : R_2^-\tilde{\theta}] = 2^{(1/2)|G|}.$$

But

$$[R_2\tilde{\theta} : I_2^-] = 2,$$

from Lemma 6.22. Putting everything together, we obtain

$$[R_2^- : I_2^-] = \text{2-part of } h^-,$$

as desired.

Finally, we consider $q = p$. The main problem is that θ has p^n in its denominator. Let $\tilde{\theta} = \theta - \frac{1}{2}N$ be as above. Suppose $x = \sum_{(b,p)=1} x_b\sigma_b \in R_p$. Then $x\tilde{\theta} \in R_p^- \Leftrightarrow x\theta \in R_p$. We have

$$x\theta = \frac{1}{p^n}\sum_c \sum_a a x_{ac}\sigma_c;$$

hence $x\theta \in R_p \Leftrightarrow \sum_a ax_{ac} \equiv 0 \pmod{p^n}$ for all c with $(c, p) = 1$. But

$$\sum_a ax_{ac} \equiv c^{-1} \sum_a acx_{ac} \equiv c^{-1} \sum_a ax_a \pmod{p^n},$$

so we only need $\sum ax_a \equiv 0 \pmod{p^n}$. It follows easily that $(x - b)\theta \in R_p$ for exactly one integer $b \pmod{p^n}$. Therefore

$$[R_p\tilde{\theta} : R_p\tilde{\theta} \cap R_p^-] = p^n.$$

But

$$R_p\tilde{\theta} = R_p^-\tilde{\theta} = R_p^-(\theta - \tfrac{1}{2}N) = R_p^-\theta, \quad \text{since } \frac{1-J}{2}N = 0.$$

Therefore

$$R_p\tilde{\theta} \cap R_p^- = R_p^-\theta \cap R_p^- \subseteq R_p\theta \cap R_p^- = I_p^-.$$

If $x\theta \in R_p^-$, then $x\theta = [(1 - J)/2]x\theta \in R_p^-\theta$; hence $R_p\theta \cap R_p^- \subseteq R_p^-\theta \cap R_p^-$. Therefore

$$I_p^- = R_p\tilde{\theta} \cap R_p^-.$$

From the above,

$$[R_p^-\theta : I_p^-] = p^n.$$

Let

$$A: R_p^- \to R_p^-, \qquad x \mapsto p^n\theta x.$$

Then

$$\begin{aligned}[R_p^- : p^nR_p^-\theta] &= p\text{-part of } \det(A) \\ &= p\text{-part of } p^{(n/2)|G|} \prod B_{1,\bar{\chi}} \\ &= p^{(n/2)|G|}\left(\frac{1}{p^n}\right)(p\text{-part of } h^-).\end{aligned}$$

But $[R_p^-\theta : p^nR_p^-\theta] = p^{(n/2)|G|}$, so

$$[R_p^- : I_p^-] = p\text{-part of } h^-.$$

This completes the proof of Theorems 6.19 and 6.21. □ □

The formula $[R^- : I^-] = h^-$ may be regarded as an algebraic interpretation of the class number formula. It should be considered as being of a similar nature to the formula $[E^+ : C^+] = h^+$, which will be proved in Chapter 8.

The natural question arises: is there an isomorphism of G-modules

$$R^-/I^- \simeq C/C^+?$$

(C = class group, C^+ = class group of the real subfield). After all, both sides have the same order. In general, there is not such an isomorphism. Let

$p = 4027$. The class group C_1 of $\mathbb{Q}(\sqrt{-4027})$ is $\mathbb{Z}/3\mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z}$. Note that since $1 + J$ is the norm to $\mathbb{Q}$, J acts by inversion on the ideal class group. Therefore $C_1^- = C_1$ and $C_1^+ = 1$. Since $\mathbb{Q}(\zeta_p)/\mathbb{Q}(\sqrt{-p})$ is totally ramified (at p), it follows from class field theory that the norm map on the ideal class groups is surjective. Suppose $R^-/I^- \simeq C/C^+$. Since R^-/I^- is cyclic over R, generated by $1 - J$, a similar statement holds for C/C^+; so there exists $c \in C$ such that $C/C^+ = cR \bmod C^+$. Therefore $C_1 = \mathrm{Norm}(C) = \mathrm{Norm}(C/C^+)$ (since $\mathrm{Norm}(C^+) \subseteq C_1^+ = 1$) is generated by $c_1 = \mathrm{Norm}(c)$ over R. If $\sigma \in \mathrm{Gal}(\mathbb{Q}(\zeta_p)/\mathbb{Q})$ then $\sigma | \mathbb{Q}(\sqrt{-p}) = 1$ or J. Therefore $c_1^\sigma = c_1^{\pm 1}$. It follows that $c_1 R$ is the subgroup generated by c_1, hence $c_1 R \neq C_1 \simeq \mathbb{Z}/3\mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z}$. Therefore R^-/I^- is not isomorphic to C/C^+.

However, we may hope for less. Let A be the p-Sylow subgroup of the ideal class group of $\mathbb{Q}(\zeta_{p^n})$, so $A = A^+ \oplus A^-$. Then is there a G-isomorphism

$$R_p^-/I_p^- \simeq A^-?$$

Again, both sides have the same order. We shall show in Chapter 10 that if $p \nmid h^+(\mathbb{Q}(\zeta_p))$ then the above holds, so A^- is cyclic (i.e., generated by one element) as a module over the group ring in that case.

§6.5. Fermat's Last Theorem

Theorem 6.23. *Suppose p is prime and suppose the index of irregularity of p (= the number of Bernoulli numbers divisible by p) satisfies $i(p) < \sqrt{p} - 2$. Then*

$$X^p + Y^p = Z^p, \qquad (XYZ, p) = 1,$$

has no integer solutions.

Remark. This theorem was proved by Eichler under the assumption that the p-rank of the minus part of the ideal class group is less than $\sqrt{p} - 2$. It was noticed (independently) by Brückner, Iwasawa, and Skula that it is possible to use $i(p)$ instead. This yields a stronger theorem. Ribet's theorem shows that $i(p) \leq$ rank, but since possibly some component A_i could have rank greater than 1, we could have strict inequality. Also, it is easier to compute $i(p)$.

Up to 4000000, the large value of $i(p)$ is 7, and probability arguments indicate that we should have $i(p) = O(\log p/\log\log p)$. Therefore, the theorem gave perhaps the best evidence before Wiles for the first case of Fermat's Last Theorem. In fact, the probability that $i(p) > \sqrt{p} - 2$ is

$$\sum_{k > \sqrt{p} - 2} e^{-1/2} \frac{(\frac{1}{2})^k}{k!}$$

(see the discussion following Theorem 5.17), which is easily seen to be at most $(1/2^k)(1/k!)$, with $k = [\sqrt{p}] - 1$. Since the first case of Fermat's Last

Theorem was known for all $p < 6 \times 10^9$, it sufficed to consider larger p. Therefore, the total number of expected exceptions to the first case of Fermat's Last Theorem was at most

$$\sum_{p > 6 \times 10^9} \left(\frac{1}{2}\right)^{[\sqrt{p}]-1} \frac{1}{([\sqrt{p}] - 1)!},$$

which is less than $10^{-300000}$. Using the more recent estimate $p < 7.57 \times 10^{17}$ yields a number that is even smaller.

Proof of Theorem 6.23. Let $\zeta = \zeta_p$. As in Chapter 1, we assume we have a solution and obtain

$$(x + \zeta^i y) = C_i^p, \qquad i = 0, \ldots, p-1,$$

where C_i is an ideal of $\mathbb{Q}(\zeta_p)$. Let C be the subgroup of the ideal class group generated by $C_1, \ldots, C_{p-1}$. Then C is an elementary p-group, so $\mathbb{Z}_p[G]$ acts on C, where $G = \mathrm{Gal}(\mathbb{Q}(\zeta_p)/\mathbb{Q})$. Since $C_1^{\sigma_a} = C_a$, C_1 generates C over the group ring, and

$$\bigoplus_i \langle \varepsilon_i C_1 \rangle = C$$

($\langle x \rangle$ denotes the cyclic subgroup generated by x). Also,

$$\bigoplus_{i \text{ odd}} \langle \varepsilon_i C_1 \rangle = C^-.$$

Therefore

$$\begin{aligned} p\text{-rank } C^- &= \#\{\varepsilon_i C_1 \neq 0,\, i \text{ odd}\} \\ &\leq \#\{A_i \neq 0,\, i \text{ odd}\} \\ &\leq i(p) \quad \text{(by Herbrand's theorem)} \\ &< \sqrt{p} - 2 \quad \text{(by assumption).} \end{aligned}$$

This type of inequality can prove useful whenever one is working with a group, such as C, which is cyclic as a module over the group ring, since in essence we have reversed the inequality "rank $\geq i(p)$."

We now proceed with the proof of the theorem, following Eichler's argument. We may assume $p > 3$. Let $r = [\sqrt{p}] - 1$. Consider the set of all products $C_1^{b_1} \cdots C_r^{b_r}$ with $0 \leq b_i < p$. The number of such products is $p^r > p^{\mathrm{rank}(C^-)} = |C^-|$. Therefore, two of them must agree in their C^--components, so we may divide and obtain

$$\prod_{i=1}^{r} C_i^{a_i} \in C^+, \quad \text{with } -p < a_i < p,$$

and some a_i is nonzero. Therefore

$$\prod C_i^{a_i} = (\rho) S,$$

with $\rho \in \mathbb{Q}(\zeta_p)$ and S an ideal with $\bar{S} = S$, so

$$\left(\prod_{i=1}^{r} (x + \zeta^i y)^{a_i}\right) = (\rho^p)S^p.$$

Since all the C_i's are prime to p, we may assume ρ and S are prime to p. The above implies that S^p is principal in $\mathbb{Q}(\zeta_p)$. Since the ideal class group of $\mathbb{Q}(\zeta_p)^+$ injects into that of $\mathbb{Q}(\zeta_p)$, by Theorem 4.14, S^p is principal in $\mathbb{Q}(\zeta_p)^+$: $S^p = (\alpha)$, with $\bar{\alpha} = \alpha$. Since any unit of $\mathbb{Q}(\zeta_p)$ is a root of unity times a real unit, we obtain

$$\prod_{i=1}^{r} (x + \zeta^i y)^{a_i} = \zeta^\mu \varepsilon \alpha \rho^p, \quad \text{with } \mu \in \mathbb{Z} \text{ and } \varepsilon \text{ a real unit.}$$

Therefore

$$\prod_{i=1}^{r} (x + \zeta^{-i} y)^{a_i} = \zeta^{-\mu} \varepsilon \alpha \bar{\rho}^p.$$

By Lemma 1.8, $\rho^p \equiv \bar{\rho}^p \equiv$ rational integer (mod p), so

$$\prod_{i=1}^{r} \left(\frac{x + \zeta^i y}{x + \zeta^{-i} y}\right)^{a_i} \equiv \zeta^{2\mu} \pmod{p},$$

and

$$\prod_{i=1}^{r} \left(\frac{x + \zeta^i y}{y + \zeta^i x}\right)^{a_i} \equiv \zeta^{v} \pmod{p},$$

with $v \equiv 2\mu - \sum i a_i \pmod{p}$, $v \geq 0$. Let

$$x_i = \begin{cases} y, & \text{if } a_i < 0 \\ x, & \text{if } a_i \geq 0, \end{cases} \qquad y_i = \begin{cases} x, & \text{if } a_i < 0 \\ y, & \text{if } a_i \geq 0, \end{cases}$$

$$F(T) = \prod (x_i + T^i y_i)^{|a_i|}, \qquad G(T) = \prod (y_i + T^i x_i)^{|a_i|}.$$

Then F yields the numerator and G yields the denominator of the above expression, so

$$F(\zeta) \equiv \zeta^v G(\zeta) \pmod{p},$$

so

$$F(\zeta) = \zeta^v G(\zeta) + pK(\zeta), \quad \text{for some } K(T) \in \mathbb{Z}[T].$$

It follows that $F(T) = T^v G(T) + pK(T) + (1 + T + \cdots + T^{p-1})H(T)$ for some polynomial $H(T) \in \mathbb{Q}[T]$, but since everything else has integral coefficients, $H(T) \in \mathbb{Z}[T]$. Multiply by $1 - T$, differentiate with respect to T, set $X = \zeta$, and reduce mod p. Then

$$(1 - \zeta)F'(\zeta) - F(\zeta) \equiv (1 - \zeta)\zeta^v G'(\zeta) - \zeta^v G(\zeta) + v(1 - \zeta)\zeta^{v-1} G(\zeta) \pmod{p}.$$

Dividing by $F(\zeta) \equiv \zeta^v G(\zeta) \pmod{p}$, we find that

$$(1-\zeta)\frac{F'(\zeta)}{F(\zeta)} - 1 \equiv (1-\zeta)\frac{G'(\zeta)}{G(\zeta)} - 1 + v(1-\zeta)\zeta^{-1} \pmod{p}.$$

This is essentially the same as what we would have obtained if we could have taken the logarithmic derivative of $F(\zeta) \equiv \zeta^v G(\zeta)$ with respect to ζ. The above may be rewritten as

$$(1-\zeta)\sum_{i=1}^{r} i|a_i|\zeta^{i-1}\frac{y_i}{x_i+\zeta^i y_i} \equiv (1-\zeta)\sum_{i=1}^{r} i|a_i|\zeta^{i-1}\frac{x_i}{y_i+\zeta^i x_i} + v(1-\zeta)\zeta^{-1},$$

which is the same as

$$(1-\zeta)\sum_{i=1}^{r} ia_i\zeta^i\left(\frac{y}{x+\zeta^i y} - \frac{x}{y+\zeta^i x}\right) \equiv v(1-\zeta) \pmod{p}.$$

Multiple by $\prod_{i=1}^{r}(x+\zeta^i y)(y+\zeta^i x)$, which is a polynomial of degree $r(r+1)$ in ζ. Let i_0 be the index of the first nonzero a_i. One the left we obtain a polynomial in ζ of degree

$$1 + i_0 + r(r+1) - 2i_0 = 1 + r(r+1) - i_0$$

with leading coefficient $i_0 a_{i_0}(x^2 - y^2)x^r y^r$. On the right we have a polynomial in ζ of degree $1 + r(r+1)$ with leading coefficient $-x^r y^r v$. But

$$1 + r(r+1) < 1 + (\sqrt{p}-1)(\sqrt{p}) < p - 1,$$

so we have a polynomial in ζ of degree less than $p-1$. By Lemma 1.9, corresponding coefficients are congruent mod p. It follows that $v \equiv 0 \pmod{p}$, so the right-hand side vanishes. The leading coefficient on the left now must vanish (mod p), so $x^2 \equiv y^2$, hence $x \equiv \pm y \pmod{p}$.

Interchanging y and z, we may also obtain $x \equiv \pm z \pmod{p}$, so

$$\pm x^p \pm x^p \equiv \pm x^p,$$

which is impossible for $p > 3$. This completes the proof. □

Notes

A good reference for much of this chapter is Coates [7]. See also the new edition of Ireland–Rosen [2].

For more on Gauss sums, see Ireland–Rosen [2] and Weil [1], [2], [3], [4].

Stickelberger's theorem was proved for $\mathbb{Q}(\zeta_p)$ by Kummer [1], and in general by Stickelberger [1]. For a proof which does not use Gauss sums, see Fröhlich [4].

Schmidt [6] defined analogues of the Stickelberger ideal for ray class groups of cyclotomic fields. He showed that the ideal annihilates the corresponding ray class group and calculated the index in terms of the ray class number.

There is a beautiful relation between Gauss sums and the p-adic Γ-function. See Gross–Koblitz [1], Lang [5], and Koblitz [4].

There are also analogues of Stickelberger's theorem for totally real fields (G. Gras [10], Oriat [3], Wiles [5]), for group rings (McCulloh [1], [2]), and K-groups (Coates–Sinnott [1], [3]). For another extension, using Hecke characters, see Iwasawa [28].

Theorem 6.19 is due to Iwasawa [11]. The analogous formula for $\mathbb{Q}(\zeta_n)$ and some other abelian fields has been proved by Sinnott [1], [2], [3].

For a detailed study of the Stickelberger ideal, see the papers of Skula. Kučera [4] determined explicit bases for the Stickelberger ideals for arbitrary cyclotomic fields and was thus able to give an easier proof of Sinnott's result in this case. For the general theory of Stickelberger ideals, see the papers of Kubert–Lang.

Jacobi sums can be used to define Hecke characters. See Weil [2], [4], Coleman–McCallum [1], Kubert [3], Kubert–Lichtenbaum [1], and Miki [8], [12].

Theorem 6.23 has been improved slightly by Uehara [2].

Exercises

6.1. Let $\mathbb{F}$ be a finite field. Show that the characters $\psi_c(x) = \psi(cx)$ for $c \in \mathbb{F}$ are distinct. Conclude that all additive characters of $\mathbb{F}$ are of this form.

6.2. Suppose the characters χ_i are multiplicative characters of $\mathbb{F}^\times$ and that $\chi_i^m = 1$ for all i. Let $e_i \in \mathbb{Z}$. Show that

$$\prod_i g(\chi_i)^{e_i} \in \mathbb{Q}(\zeta_m) \Leftrightarrow \prod \chi_i^{e_i}(a) = 1 \quad \text{for all } a \in (\mathbb{Z}/p\mathbb{Z})^\times.$$

6.3. Let $p \equiv 3 \pmod 4$ be prime, let R denote the number of quadratic residues mod p in the interval $(0, p/2)$, and let N denote the number of nonresidues in this interval. Use Stickelberger's theorem to show that $R - N$ annihilates the ideal class group of $\mathbb{Q}(\sqrt{-p})$. (Historically, this was known before the class number formula) (*Hint*: Exercise 4.5).

6.4. Let $\mathbb{E}/\mathbb{F}$ be an extension of finite fields, $[\mathbb{E} : \mathbb{F}] = n$, and let N be the norm for this extension. Let $\psi_{\mathbb{E}}$ and $\psi_{\mathbb{F}}$ be the additive characters. Let $\chi_{\mathbb{F}}$ be a multiplicative character of $\mathbb{F}$ and let $\chi_{\mathbb{E}} = \chi_{\mathbb{F}} \circ N$, a multiplicative character of $\mathbb{E}$. Let

$$R = \frac{g(\chi_{\mathbb{E}})}{g(\chi_{\mathbb{F}})^n}$$

(a) If $\chi_{\mathbb{F}}^m = 1$, show that $R \in \mathbb{Q}(\zeta_m)$.
(b) Show that R is a unit.
(c) Show that R has absolute value 1, hence is a root of 1.
(d) Use the Remark following Proposition 6.13 to show that R is congruent to 1 modulo primes above p (= characteristic of $\mathbb{F}$).
(e) (Davenport–Hasse) Conclude that for $p > 2$, $g(\chi_{\mathbb{E}}) = g(\chi_{\mathbb{F}})^n$ (this also works for $p = 2$; see the original paper by Davenport and Hasse. For a different proof, see Weil [1]).

(f) Show that if $\chi_{\mathbb{F}}$ has order exactly d then $\chi_{\mathbb{E}}$ has order exactly d.
(g) Suppose $\mathbb{F}$ has q elements and $d|q-1$. Show that the number of solutions in projective space over $\mathbb{E}$ of $X^d + Y^d = Z^d$ is $1 + q^n - \sum_{a,b=1,a+b\neq d}^{d-1} J(\chi_{\mathbb{F}}^a, \chi_{\mathbb{F}}^b)^n$.
(h) Let α be a root of the polynomial in the numerator of the zeta function for $X^d + Y^d = Z^d$ over $\mathbb{Z}/p\mathbb{Z}$. Suppose $[\mathbb{F} : \mathbb{Z}/p\mathbb{Z}] = e$. Show that $\alpha^e = J(\chi_{\mathbb{F}}^a, \chi_{\mathbb{F}}^b)$ for some a, b.

6.5. Show that for each positive integer d, the equation $X^d + Y^d = Z^d$ has nontrivial p-adic solutions for all p.

6.6. (a) Use the Brauer–Siegel theorem (or Theorem 4.19) to show that the index of irregularity satisfies $i(p) \leq p/4 + o(p)$.
(b) The probability that $i(p) = k$ is $e^{-1/2}(\frac{1}{2})^k/k!$ (see the discussion after Theorem 5.17). Let x be such that the expected number of $p \leq x$ with $i(p) = k$ is 1. Show that $\log x$ is approximately $k \log k$. Assuming that x is approximately equal to the first p with $i(p) = k$, conclude that we should have $i(p) = O(\log p/\log\log p)$. This gives a fairly accurate estimate. The first p with $i(p) = 5$ is 78233, and $\log p/\log\log p = 4.65$. The first p with $i(p) = 7$ is 3238481, which gives $\log p/\log\log p = 5.54$. This corresponds to the first occurrence of $i(p) = 7$ being earlier than expected.